N-body dynamics of Topology-Generated Potential:

Earnshaw violation, chaos suppression, and the EFT origin of Yukawa coupling

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Abstract

We derive and verify numerically the complete N-body dynamics arising from the TGP scalar-field Lagrangian $\mathcal{L}[\Phi]$ with quartic self-interaction. The potential decomposes as $V = V_2 + V_3^{\text{irr}}$, where V_2 is a closed-form pairwise interaction (gradient + repulsive + confining) and V_3 is an irreducible three-body term computed via a Feynman two-parameter integral.

Seven principal results emerge. (1) TGP breaks the Earnshaw theorem: equilateral three-body configurations admit stable static equilibria with all normal modes positive ($\omega^2 > 0$), whereas Newton gives marginal modes ($\omega^2 \approx 0$) for all tested parameter pairs (β, C). (2) TGP suppresses three-body chaos by 35–45% (measured via the maximal Lyapunov exponent) for $\beta \gtrsim 0.025$ at initial conditions involving close encounters; the effect is IC-dependent and enhanced by V_3. (3) The Yukawa source coupling C_{eff} is derived from EFT projection of the classical topological defect. (4) Classical soliton interaction is *not* Yukawa (oscillatory, power-law decay), establishing that the Yukawa law requires the EFT mass gap. (5) The irreducible three-body integral admits a semi-analytic multipole representation with controlled accuracy ($< 0.01\%$ at $L_{\max} = 15$). (6) The ratio $|V_3/V_2|$ is mapped across the dimensionless parameter $t = m_{\text{sp}}\, d$, showing perturbative behaviour ($\sim 1\%$) in the Lyapunov regime and exponential suppression for $t > 3$. (7) Closed-form expressions for the equilibrium distances d_{rep} and d_{well}, the radial normal-mode frequency ω^2_{rad}, and the TGP Hill sphere radius are derived and verified; TGP generically expands the Hill sphere relative to Newton. (8) For unequal masses, the phase diagram is governed by a critical coupling $\beta_{\text{crit}} = \frac{3}{2}\sqrt{\gamma(C_1 + C_2)}$; the escape velocity and a three-body breathing mode are derived in closed form.

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1 Introduction

The gravitational N-body problem has been a cornerstone of mathematical physics since Newton. In the restricted three-body case, the problem exhibits chaotic trajectories, non-integrability (Bruns theorem [Bruns, 1887]), and the impossibility of stable static equilibria (Earnshaw's theorem [Earnshaw, 1842]). Modern treatments range from the classical monographs of Valtonen and Karttunen [2006] and Heggie and Hut [2003] to statistical approaches for chaotic outcomes [Stone and Leigh, 2019, Manwadkar et al., 2021]. The Burrau problem [Burrau, 1913, Szebehely and Peters, 1967] remains a standard benchmark for numerical integrators.

In modified-gravity theories, N-body dynamics are altered by additional scalar or vector degrees of freedom. Cosmological N-body codes incorporating scalar-field modifications include MG-GADGET [Puchwein et al., 2013], ECOSMOG [Li et al., 2012], and ISIS [Llinares et al., 2014]; a systematic comparison was carried out by Winther et al. [2015]. These codes focus on large-scale structure ($\gtrsim$ Mpc), and screening mechanisms (chameleon [Khoury and Weltman, 2004], Vainshtein [Vainshtein, 1972]) play a central role. At the fundamental level, scalar-tensor theories [Damour and Esposito-Farèse, 1992] introduce a Yukawa correction to the Newtonian potential, constrained by fifth-force experiments [Adelberger et al., 2003, Murata and Tanaka, 2015] and solar-system tests [Will, 2014, Bertotti et al., 2003].

A separate line of research concerns *few-body* systems with short-range (Yukawa-type) interactions, where three-body forces can produce qualitatively new phenomena. The Efimov effect [Efimov, 1970] — an infinite tower of three-body bound states near a two-body resonance — illustrates how irreducible three-body physics can dominate; see Braaten and Hammer [2006] and Naidon and Endo [2017] for reviews. While the Efimov effect is typically studied in quantum mechanics, the underlying principle — that non-additive three-body forces qualitatively alter the spectrum — is relevant to any system with short-range interactions.

The Topology-Generated Potential (TGP) [Serafin, 2026] is a scalar-field theory in which space is emergent from a discrete substrate. The field Lagrangian naturally produces a quartic self-interaction that, upon expansion around solitonic sources, yields: (i) a pairwise potential V_2 combining Newtonian attraction, a repulsive barrier, and short-range confinement; (ii) an irreducible three-body potential V_3 arising from the Φ^4 vertex. This paper derives the complete N-body dynamics from the TGP Lagrangian and presents eight principal results, each verified by a dedicated regression suite (59/59 PASS, 102 active scripts).

The paper is organized as follows. Section 2 derives the equations of motion. Section 3 establishes Earnshaw violation (Result 1). Section 4 quantifies chaos suppression (Result 2). Section 5 derives the EFT Yukawa coupling (Result 3). Section 6 contrasts classical soliton interaction with the Yukawa law (Result 4). Sections 7–8 analyze the three-body integral and its regime map (Results 5–6). Section 9 derives analytical equilibria and the Hill criterion (Result 7). Section 11 treats unequal masses and the phase diagram (Result 8). Section 12 summarizes the deductive chain.

2 Equations of motion from TGP Lagrangian

2.1 Setup

We consider N point sources with positions $\mathbf{x}_i \in \mathbb{R}^3$ and source strengths $C_i > 0$ serving as both gravitational "charge" and inertial mass ($m_i \equiv C_i$). The TGP Lagrangian yields a total potential $V = V_2 + V_3$ with the following closed-form expressions.

2.2 Pairwise potential V_2

For bodies $i \neq j$ at separation $d_{ij} = |\mathbf{x}_i - \mathbf{x}_j|$,

$$\boxed{V_{2,ij}(d_{ij}) = \underbrace{-\frac{4\pi\, C_i C_j}{d_{ij}}}_{\text{gradient (attractive)}} + \underbrace{\frac{8\pi\,\beta\, C_i C_j}{d_{ij}^2}}_{\text{repulsive } (\beta\text{-term})} - \underbrace{\frac{12\pi\,\gamma\, C_i C_j\,(C_i + C_j)}{d_{ij}^3}}_{\text{confining } (\gamma\text{-term})},} \tag{1}$$

with total $V_2 = \sum_{i<j} V_{2,ij}$. The three terms arise from the $(\nabla\Phi)^2$, Φ^3, and Φ^4 sectors of the TGP action, respectively. The hierarchy $1/d \gg 1/d^2 \gg 1/d^3$ at large separation ensures Newtonian behaviour at long range; at short distances the β-repulsion creates a barrier d_{rep} (when it exists) and the γ-confinement creates a well d_{well}.

2.3 Irreducible three-body potential V_3

For each unordered triplet $\{i, j, k\}$, the Φ^4 vertex contributes

$$\boxed{V_{3,ijk} = -6\,\gamma\, C_i C_j C_k\; I_Y(d_{ij}, d_{ik}, d_{jk};\, m_{\text{sp}}),} \tag{2}$$

where I_Y is the triple-overlap integral in the Feynman (Schwinger-parameter) representation:

$$I_Y = 2\int_{\Delta_2} \Delta^{-3/2}\, K_0\!\Big(m_{\text{sp}}\sqrt{Q/\Delta}\Big)\, \mathrm{d}\alpha, \tag{3}$$

with $\Delta = \alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3$, $Q = \alpha_2 d_{ij}^2 + \alpha_1 d_{ik}^2 + \alpha_3 d_{jk}^2$, the integral being over the standard 2-simplex $\Delta_2 = \{\alpha \geq 0,\ \alpha_1 + \alpha_2 + \alpha_3 = 1\}$. The screening mass is

$$m_{\rm sp} = \sqrt{3\gamma - 2\beta}. \tag{4}$$

In the unscreened (Coulomb) limit $m_{\rm sp} \to 0$: $I \to 8\pi^2/(d_{ij} + d_{ik} + d_{jk})$.

2.4 Equations of motion

With $V = V_2 + \sum_{i<j<k} V_{3,ijk}$, the Euler–Lagrange equations give

$$C_i\,\ddot{\mathbf{x}}_i = \mathbf{F}_i = -\nabla_{\mathbf{x}_i} V, \tag{5}$$

with exact translational invariance ($\sum_i \mathbf{F}_i = 0$), $SO(3)$ rotational invariance ($\sum_i \mathbf{x}_i \times \mathbf{F}_i = 0$), and Hamiltonian structure ($H = T + V$ conserved).

Remark. V_2 and V_3 are the *only* terms arising from the standard TGP Φ^4 Lagrangian for point-source superposition; there is no free structural parameter beyond $(\beta, \gamma, \{C_i\})$.

3 Result 1: Breaking Earnshaw's theorem

Result 1 (Earnshaw violation)**.** *For equilateral three-body configurations at the confining equilibrium $d_{\rm well}$, TGP produces* ***stable*** *static equilibria (all $\omega^2 > 0$), while Newton yields marginal modes ($\omega^2 \approx 0$) for all tested (β, C) pairs.*

3.1 Background

Earnshaw's theorem [Earnshaw, 1842] states that no static equilibrium of point masses under an inverse-square-law force can be stable in all directions. In TGP, the higher-order terms ($1/d^2$ repulsion, $1/d^3$ confinement) create an effective well $d_{\rm well}$ where $V_2'(d_{\rm well}) = 0$ with $V_2''(d_{\rm well}) > 0$. The question is whether the full $3N$-dimensional Hessian $\partial^2 V/\partial x_{i\alpha}\partial x_{j\beta}$ has all non-translational eigenvalues positive.

3.2 Normal-mode analysis

We compute the mass-weighted dynamical matrix $D = M^{-1/2}\,H\,M^{-1/2}$ where H is the $3N \times 3N$ Hessian and $M = \mathrm{diag}(C_1, C_1, C_1, C_2, \ldots)$. Normal-mode frequencies satisfy $\omega_k^2 =$ eigenvalues of D, with translational zero modes removed.

3.3 Results

For an equilateral triangle ($N = 3$) on $d_{\rm well}$, scanning $\beta \in [0.5, 5.0]$ and $C \in [0.05, 0.50]$:

Mode type	**Newton ω^2**	**TGP ω^2**	**Character**
Breathing (radial)	~ 0 (marginal)	0.0083	Radial-fraction = 1.0
Mixed $\times 2$ (C_3 degen.)	~ 0	0.0042	Tangential + radial

- Newton: **marginal** in 11/11 (β, C) points (Earnshaw).
- TGP: **stable** in 11/11 points — all physical modes $\omega^2 > 0$.
- Effect of V_3: mildly destabilizes 1 of 3 modes (toward marginal), but 2 modes remain stable.
- 5-gon ($N = 5$): TGP creates a *saddle* ($\omega^2 = -20.7$ tangential, $+24.7$ radial) — stabilization is **geometry-specific**, not universal.

Significance. TGP is the only known modification of gravity that breaks Earnshaw for static N-body configurations *without rotation or external fields* — purely from $V(\Phi)$.

4 Result 2: Chaos suppression in the three-body problem

Prediction 1 (Falsifiable). *For $\beta \gtrsim 0.025$ and initial conditions with close encounters, TGP suppresses the maximal Lyapunov exponent by 35–45%*:

$$\frac{\lambda_{\max}^{\mathrm{TGP}}}{\lambda_{\max}^{\mathrm{Newton}}} \approx 0.55\text{–}0.65. \tag{6}$$

4.1 Methodology

We set $G_{\mathrm{Newton}} = 4\pi$ so that the leading TGP term $V_{\mathrm{grad}} = -4\pi C_i C_j/d$ reproduces Newton in the limit $\beta \to 0$. Kinetic energy is matched so that $H_{\mathrm{TGP}} = H_{\mathrm{Newton}}$ on the same configuration (Burrau pythagorean IC [Burrau, 1913, Szebehely and Peters, 1967] with $v = 0$; energy matching via velocity scaling or random velocity generation with fixed RNG seed for reproducibility).

The maximal Lyapunov exponent $\lambda_{\max}$ is computed via the Benettin algorithm [Benettin et al., 1980]:

1. Symplectic leapfrog (Velocity Verlet) for the base trajectory.
2. Coupled tangent map with the full analytical Jacobian $J_{i\alpha,j\beta} = \partial a_{i,\alpha}/\partial x_{j,\beta}$ (chain rule through the Hessian of I_Y in the distance variables).
3. QR renormalization at regular intervals; accumulation of $\sum \log \|R_{kk}\|$.

For the full spectrum ($k = 6N = 18$ exponents), Hamiltonian conservation is verified: $|\sum_i \lambda_i| \sim 10^{-13}$.

4.2 Beta scan with $V_2 + V_3$ (Burrau IC)

β	ratio(V_2+V_3)	ratio(V_2)	V_3 effect	Classification
0.020	1.09	1.91	−43%	comparable
0.050	0.52	0.67	−22%	SUPPRESS
0.080	0.58	0.91	−36%	SUPPRESS
0.120	0.63	0.99	−36%	SUPPRESS
0.200	0.65	0.85	−24%	SUPPRESS
0.350	0.60	0.76	−21%	SUPPRESS

- 9/10 β-points: SUPPRESS (with V_3) vs 7/10 (V_2-only).
- $\beta_{\mathrm{crit}}(V_3) \approx 0.025$ vs $\beta_{\mathrm{crit}}(V_2) \approx 0.042$ — V_3 **lowers threshold by** $\sim 40\%$.
- Convergence: ratio stable from $t = 2$ to $t = 6$.

4.3 IC-dependence (multi-IC scan)

IC type	**SUPPRESS fraction**	**Mechanism**
Burrau (close encounters)	75%	Potential softening at small d
Equilateral (symmetric)	0%	No close encounters

The suppression mechanism is *not* the repulsive barrier d_{rep} (which does not exist for typical C values; it requires $\beta \gg 1$). Rather, TGP's higher-order terms ($1/d^2$, $1/d^3$) **effectively soften the potential** at close encounters, reducing the exponential divergence of nearby trajectories. Symmetric ICs (equilateral) never enter the close-encounter regime and thus show no suppression.

4.4 Caveats

1. Finite-time Lyapunov exponents ($t_{\mathrm{final}} \leq 6$).

2. Burrau IC with $v = 0$ requires random velocities for energy matching — orbits depend on the RNG seed (seed 200 fixed for reproducibility).
3. Leapfrog energy drift grows at long horizons; RK45 gives `success=False` at close encounters.

5 Result 3: Yukawa coupling from EFT projection of topological defect

Result 2 (Path C closure). *The Yukawa source axiom (Path B: $\delta_i = C_i\, e^{-m_{\rm sp} r}/(4\pi r)$) is* ***derived*** *from the physics of the classical TGP defect via an EFT projection.*

5.1 The problem

The classical TGP field equation for a spherical defect [cf. Manton and Sutcliffe, 2004, for soliton physics], $g^2 g'' + g(g')^2 + (2/r)\, g^2\, g' = g^2(1-g)$, admits solutions with boundary conditions $g(0) = g_0 < 1$, $g(\infty) = 1$. Linearization at $g \to 1$ gives $(\nabla^2 + 1)\delta = 0$ where $\delta = 1 - g$, yielding an **oscillatory** tail $\delta(r) \sim \sin(r)/r$ (verified numerically: 7 sign changes). This is *not* Yukawa.

5.2 Stabilization obstacle

A naive vacuum-stabilization potential $V_{\rm sb} = (\mu^2/2)(g-1)^2$ fails: the derivative $V_C'(g) = (1-g)(g^2 - \mu^2)$ is negative for all $g \in (0,1)$ when $\mu^2 > 1$, destroying the defect's turning point.

5.3 EFT resolution

The screening mass $m_{\rm sp}$ enters not through a classical Lagrangian modification, but through **loop corrections to the propagator** (EFT mass gap). The effective source coupling is the projection of the classical defect profile onto the Yukawa Green's function:

$$\boxed{C_{\rm eff} = \int_0^\infty \delta(r)\, e^{-m_{\rm sp}\, r}\, r\, \mathrm{d}r,} \tag{7}$$

where $\delta(r) = 1 - g(r)$ is the classical defect profile.

5.4 Scaling laws

Scanning over defect depth g_0 and coupling β:

$$|C_{\rm eff}| \propto (1 - g_0)^{1.02}, \qquad |C_{\rm eff}| \propto \beta^{-1.00}. \tag{8}$$

The exponents are remarkably close to unity, giving a transparent physical interpretation: deeper cores and weaker self-coupling produce larger effective Yukawa charges.

The self-consistent screening mass is $\mu^2 = 2(3\gamma - 2\beta)$, identical to Eq. (4) squared, confirming internal consistency of the EFT framework.

6 Result 4: Classical soliton interaction is not Yukawa

Result 3 (EFT mass gap required). *The classical overlap interaction of two TGP defects is oscillatory and decays as a power law ($\sim d^{-1/2}$), not exponentially. The Yukawa force law used in N-body dynamics is a* ***quantum*** *(EFT) result, not a classical one.*

6.1 Method

Two defect profiles are placed at separation d and their field overlap is integrated on a 2D cylindrical grid (ρ, z):

$$S(d) = 2\pi \iint \delta_1\, \delta_2\, \rho\, \mathrm{d}\rho\, \mathrm{d}z \qquad \text{(overlap integral)}, \tag{9}$$

$$G(d) = \iint \nabla\delta_1 \cdot \nabla\delta_2\, \mathrm{d}^3x \qquad \text{(gradient cross integral)}, \tag{10}$$

$$V_{\mathrm{cl}}(d) = G(d) + V''(1)\, S(d) \qquad \text{(classical interaction energy)}. \tag{11}$$

6.2 Results

- $\delta(d)$ oscillates (7 sign changes) — the classical defect tail is not Yukawa.
- $G(d)$ and $V''(1)\, S(d)$ cancel to $\sim 93\%$ — confirming the linearized equation $(\nabla^2 + 1)\delta = 0$.
- V_{cl} decays as $d^{-1/2}$ (power law), not $e^{-m_{\mathrm{sp}} d}$.
- $|V_{\mathrm{cl}}/V_Y|$ grows from ~ 16 to $\sim 10^8$ with distance — completely different physics.

Significance. This closes the deductive chain:

$$g(r) \xrightarrow{\text{P5: EFT projection}} C_{\mathrm{eff}} \xrightarrow{\text{Path B}} V_Y(d) \xrightarrow{\text{P0: EOM}} \mathbf{F}_i.$$

The Yukawa law is justified *only* through the EFT mass gap, not by classical field overlap.

7 Result 5: Semi-analytic multipole representation of I_Y

Using the Yukawa addition theorem, the double sum in the partial-wave expansion of I_Y collapses to a **single sum over** l:

$$\boxed{I_Y = 4\pi \sum_{l=0}^{L_{\max}} (2l+1)\, P_l(\cos\omega)\, R_l(d_{13}, d_{23}, m_{\mathrm{sp}}),} \tag{12}$$

where ω is the angle at the vertex joining edges d_{13} and d_{23}, P_l are Legendre polynomials, and R_l is a 1D radial integral computed by 3-segment Gauss–Legendre quadrature. The key simplification is the angular diagonalization: $A_{ll'}(\omega) = \frac{4\pi}{2l+1}\, \delta_{ll'}\, P_l(\cos\omega)$.

$L_{\max}$	Relative error (equilateral)	Relative error (scalene)
3	0.23%	2.1%
10	0.12%	0.22%
15	0.006%	0.04%

The Feynman 2D integral (Sec. 2) remains faster per evaluation with caching, but the multipole form is tabulable and analytically transparent.

8 Result 6: V_3/V_2 regime map

The dimensionless parameter $t = m_{\mathrm{sp}}\, d$ controls the importance of three-body forces relative to pairwise interactions.

Regime	$\lvert V_3/V_2\rvert$	**Description**
Coulomb ($t < 0.5$)	$\sim 1.3\%$	Saturates (constant); V_3 perturbative
Transition ($t \sim 1$)	$\sim 14\%$	Peak; V_3 significant
Yukawa ($t > 3$)	$< 0.2\%$	Exponentially screened

Additional findings:

- **Lyapunov regime** ($\beta = 0.02$–0.35, $m_{\rm sp}d \sim 0.3$–1.0): $|V_3/V_2| = 0.8$–1.0% — perturbative but sufficient to lower $\beta_{\rm crit}$ by 40%.
- **Geometry**: compact configurations ($d/2$) have $7\times$ larger V_3/V_2 than extended ($2d$) — close encounters amplify three-body effects.
- **C-scaling**: $V_3/V_2 \propto C$ (linear, verified to $< 2\%$ deviation) — consistent with $V_3 \propto C^3$ vs $V_2 \propto C^2$.
- **Physical TGP** ($\gamma \sim \Lambda_{\rm eff} \sim 10^{-52}$): $m_{\rm sp}d \ll 1$ — Coulomb regime applies for all macroscopic separations.

9 Result 7: Analytical equilibria, normal modes, and Hill criterion

9.1 Exact equilibrium distances

Setting $V_2'(d) = 0$ for equal-mass bodies ($C_1 = C_2 = C$) yields a quadratic in d:

$$\boxed{d^2 - 4\beta\, d + 18\gamma\, C = 0.} \tag{13}$$

The two roots give the *repulsive barrier* and the *confining well*:

$$d_{\rm rep} = 2\beta - \sqrt{4\beta^2 - 18\gamma C}, \qquad d_{\rm well} = 2\beta + \sqrt{4\beta^2 - 18\gamma C}, \tag{14}$$

existing whenever $\beta > \sqrt{9\gamma C/2}$. For unequal masses, the condition generalizes to $d^2 - 4\beta\, d + 9\gamma(C_1 + C_2) = 0$. At threshold ($4\beta^2 = 18\gamma C$), the two equilibria merge: $d_{\rm rep} = d_{\rm well} = 2\beta$. Far from threshold: $d_{\rm rep} \to 0$ and $d_{\rm well} \to 4\beta$. Verified numerically: analytical = numerical to machine precision.

9.2 Closed-form normal-mode frequency

The radial stiffness at $d_{\rm well}$ is

$$V_2''(d) = \frac{8\pi C^2}{d^3}\left[-1 + \frac{6\beta}{d} - \frac{18\gamma C}{d^2}\right]. \tag{15}$$

For an isolated pair with reduced mass $\mu = C/2$, the radial breathing frequency is

$$\boxed{\omega_{\rm rad}^2 = \frac{2\, V_2''(d_{\rm well})}{C} = \frac{16\pi C}{d_{\rm well}^3}\left[-1 + \frac{6\beta}{d_{\rm well}} - \frac{18\gamma C}{d_{\rm well}^2}\right].} \tag{16}$$

Verified against finite-difference Hessian: relative error $< 10^{-6}$.

9.3 TGP Hill sphere

The classical Hill sphere [Murray and Dermott, 1999, Hamilton and Burns, 1992] is the region around a body in which its gravity dominates over the tidal field of a more massive companion. The tidal field of a central body $C_{\rm cen}$ at distance d in TGP is

$$\Omega_{\rm tidal}^2(d) = \frac{8\pi C_{\rm cen}}{d^3}\left[1 - \frac{6\beta}{d} + \frac{18\gamma\, C_{\rm cen}}{d^2}\right]. \tag{17}$$

The β-repulsion *weakens* the tidal field (negative correction to the bracket), while γ-confinement *strengthens* it. The Hill radius follows from balancing the tidal acceleration against self-gravity of a body of mass $C_{\rm body}$:

$$\boxed{R_H^3 = \frac{4\pi\, C_{\rm body}}{|\Omega_{\rm tidal}^2(d)|}.} \tag{18}$$

Scanning $\beta \in [0.01, 2.0]$ at $d = 5$, $C_{\rm body} = 0.1$, $C_{\rm cen} = 1.0$:

β	$R_H^{\rm TGP}/R_H^{\rm Newton}$	Effect
0.01	1.002	∼same
0.10	1.017	slightly larger
0.50	1.096	10% larger
1.00	1.244	24% larger
2.00	2.924	3× larger

Result: TGP generically **expands** the Hill sphere ($R_H^{\rm TGP} > R_H^{\rm Newton}$), making satellite systems more stable against tidal disruption.

9.4 Perturbative V_3 correction to equilibria

The three-body force shifts $d_{\rm well}$ by

$$\delta d = -\frac{F_3(d_{\rm well})}{3\,V_2''(d_{\rm well})}, \tag{19}$$

where $F_3 = -\partial V_3/\partial d|_{\rm breathing}$. Numerically: $|\delta d/d_{\rm well}| < 0.2\%$ for all tested (β, C) — the correction is perturbative. The sign is positive: V_3 **shifts** $d_{\rm well}$ **outward** (slightly larger equilibrium distance).

10 Supplementary result: Closed orbits and Lagrange stability

Two classical periodic configurations were tested under TGP perturbation:

1. **Figure-8 choreography** (Chenciner–Montgomery): survives TGP to $\beta = 0.10$ ($C = 0.3$). Period shift: Newton $T = 3.115$, TGP $T = 3.283$ (+5.4%). Deformation scales as $\Delta r_{\rm max} \sim \beta^{0.65}$ (perturbative). With V_3: $V_3/V_2 = 0.82\%$.
2. **Lagrange point** L_4: position shift $\sim 10^{-5}$; stability preserved ($q_{\rm crit}$ unchanged); Poincaré sections show regular orbits. Effective-potential curvature changes by $< 10^{-4}$.

Conclusion. Periodic orbits are *robust* under TGP corrections — the perturbation is smooth and non-destructive.

11 Result 8: Unequal masses, phase diagram, escape velocity, breathing mode

Result 4 (Unequal-mass confinement and phase diagram)**.** *For $C_1 \neq C_2$ the force equation $F(d) = 0$ yields*

$$d^2 - 4\beta\, d + 9\gamma(C_1{+}C_2) = 0, \tag{20}$$

with roots $d_{\rm rep,well} = 2\beta \mp \sqrt{4\beta^2 - 9\gamma(C_1{+}C_2)}$.

Eq. (20) reduces to eq. (13) for $C_1{=}C_2{=}C$.

11.1 Phase diagram

Confinement exists if and only if the discriminant is positive:

$$\boxed{\beta > \beta_{\rm crit} \;=\; \tfrac{3}{2}\sqrt{\gamma\,(C_1{+}C_2)}\,, \qquad \text{equivalently} \quad C_1{+}C_2 < \tfrac{4}{9}\,\frac{\beta^2}{\gamma}\,.} \tag{21}$$

At $\beta = \beta_{\rm crit}$ the two roots coalesce: $d_{\rm rep} = d_{\rm well} = 2\beta_{\rm crit}$ (saddle-node bifurcation). For fixed C_1, the maximum mass ratio for which equilibria persist is

$$q_{\rm max} = \frac{C_{2,\rm max}}{C_1} = \frac{4\beta^2}{9\gamma\, C_1} - 1. \tag{22}$$

Numerical verification (`ex211`, grid 10×10 in $(\beta, C_{\rm sum})$): 100% agreement with analytical boundary.

11.2 Escape velocity

The binding energy at $d_{\rm well}$ is $E_{\rm bind} = |V_2(d_{\rm well})|$, so the escape velocity in the reduced-mass frame is

$$\boxed{v_{\rm esc} = \sqrt{\frac{2\,|V_2(d_{\rm well})|}{\mu}}\,, \qquad \mu = \frac{C_1 C_2}{C_1{+}C_2}\,.} \tag{23}$$

Substituting $V_2(d_{\rm well}) = -A/d_{\rm well} + B/d_{\rm well}^2 - G/d_{\rm well}^3$ gives a fully closed-form expression in $(\beta, \gamma, C_1, C_2)$. Scaling: $v_{\rm esc}$ decreases monotonically with β (deeper well at smaller β, but $d_{\rm well}$ also shrinks; net effect: $v_{\rm esc} \propto \beta^{-0.5}$ approximately).

11.3 Three-body breathing mode

For three equal masses ($C_1{=}C_2{=}C_3{=}C$) in an equilateral triangle with side d, the effective potential for the breathing coordinate is

$$U_{\rm eff}(d) \;=\; 3\,V_2(d) \;+\; V_3(d,d,d). \tag{24}$$

The reduced mass for symmetric breathing (all distances change equally) is $\mu_{\rm br} = C$, giving

$$\omega_{\rm br}^2 = \frac{U''_{\rm eff}(d_{\rm well})}{C} = \frac{3\,V_2''(d_{\rm well}) + V_3''(d_{\rm well})}{C}\,. \tag{25}$$

Without V_3, this simplifies to $\omega_{\rm br}^2 = 3\,V_2''(d_{\rm well})/C = \frac{3}{2}\,\omega_{\rm rad}^2$ (the breathing mode is **50% stiffer** than the two-body radial mode; physically, three pairs contribute to the restoring force while the breathing reduced mass is C, not $C/2$).

The V_3 correction to $\omega_{\rm br}^2$ is -1.3% (`ex211`, $\beta{=}\gamma{=}1$, $C{=}0.1$) — perturbative and consistent with the V_3/V_2 regime analysis (Result 6).

Inward barrier. The energy barrier for the breathing mode to reach $d_{\rm rep}$ from $d_{\rm well}$ is $\Delta U = U_{\rm eff}(d_{\rm rep}) - U_{\rm eff}(d_{\rm well})$. Numerically: $\Delta U/|U_{\rm eff}(d_{\rm well})| \approx 9.4$ for $\beta{=}1$, $C{=}0.1$ — the inward barrier is roughly ten times the binding energy, confirming that the repulsive wall at $d_{\rm rep}$ is highly effective.

12 Summary and deductive chain

The complete deductive chain from the TGP field Lagrangian to N-body dynamics is:

Layer	Content
Field equation	$K_{\rm sub}(g)\,\Box g + \ldots = V'(g) \quad (K_{\rm sub} = g^2)$
Defect profile	$g(r)$: ODE solution, oscillatory tail $\sin(r)/r$
EFT projection (P5)	$C_{\rm eff} = \int \delta\, e^{-m_{\rm sp} r}\, r\, {\rm d}r$
Yukawa source (Path B)	$\delta_i = C_{\rm eff}\, e^{-m_{\rm sp} r}/(4\pi r)$
V_2 (pairwise)	Eq. (1): gradient + repulsion + confinement
V_3 (three-body)	Eq. (2): Feynman integral I_Y
N-body EOM	$C_i \ddot{\mathbf{x}}_i = -\nabla_{\mathbf{x}_i}(V_2 + V_3)$
Novel results from this chain:	
Result 1	Earnshaw violation (Sec. 3)
Result 2	Chaos suppression 35–45% (Sec. 4)
Result 3	$C_{\rm eff}$ from EFT projection (Sec. 5)
Result 4	Classical $\neq$ Yukawa (Sec. 6)
Result 5	I_Y multipole form (Sec. 7)
Result 6	V_3/V_2 regime map (Sec. 8)
Result 7	Analytical equilibria + Hill sphere (Sec. 9)
Result 8	Phase diagram, $v_{\rm esc}$, breathing mode (Sec. 11)

All results are supported by a regression suite of 59 numerical tests (102 active example scripts) passing consistently. The implementation is available in the `nbody/` package of TGP v1.

Context. The pairwise potential V_2 connects to the three regimes of TGP (gradient, repulsive, confining) identified in the parent theory [Serafin, 2026]. The N-body dynamics presented here are *exact* consequences of the field Lagrangian (no tunable parameters beyond β, γ, C_i), distinguishing TGP from phenomenological modified-gravity N-body approaches [Puchwein et al., 2013, Li et al., 2012, Winther et al., 2015]. The Earnshaw violation (Result 1) and chaos suppression (Result 2) are qualitatively new predictions with no counterpart in Newtonian gravity; the EFT derivation of $C_{\rm eff}$ (Result 3) bridges the gap between the topological defect (UV) and the Yukawa source (IR).

Outlook. Future work includes: (i) extending to relativistic corrections using the TGP metric ($\gamma_{\rm PPN} = \beta_{\rm PPN} = 1$, per Will 2014); (ii) cosmological N-body simulations incorporating scale-dependent $G_{\rm eff}(k, a)$; (iii) exploring the Efimov-like tower that may arise from V_3 near the two-body resonance at $\beta_{\rm crit}$ [Efimov, 1970, Braaten and Hammer, 2006]; (iv) applying the analytical equilibria (Result 7) and Hill criterion to exoplanetary stability [Murray and Dermott, 1999, Hamilton and Burns, 1992].

Reproducibility. All numerical results are reproduced by `python reproduce_all.py --quick` (10/10 stages PASS in ∼100 s on a standard laptop). See `README.md` for the full script manifest and the supplementary material for data files.

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